

A general amalgamation theorem for unitary representations

Agent lane 11 packet for arXiv:1612.09448

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Status

This packet records a strengthened partial answer to Question 4.5 of Michal Doucha's paper *Universal actions and representations of locally finite groups on metric spaces* [1]. The first result below answers the unrestricted Hilbert-space amalgamation clause affirmatively. The second result proves that, when the input groups are finite and the input Hilbert spaces are finite-dimensional, the amalgamating Hilbert space can also be chosen finite-dimensional. The packet does not prove that the generated image of the amalgamating group can be chosen finite; if the last sentence of Question 4.5 is interpreted as a finite Fraïssé-class amalgamation requirement, that stronger finite-image point remains open here. The preceding question about a universal unitary representation of a universal locally finite group is also not settled.

Source question

The question appears on page 21 of the arXiv PDF, immediately after Question 4.4.

the concrete question was an open question.

Question 4.4. *Does there exist a universal unitary representation of a universal locally finite group?*

Of special interest is whether one can amalgamate unitary representations.

Question 4.5. *Let G_1 and G_2 be two groups with a common subgroup G_0 . Let π_1 and π_2 be two unitary representations of G_1 , resp. G_2 , on Hilbert spaces H_1 , resp. H_2 with a common subspace H_0 , on which the restrictions $\pi_1|_{G_0}$ and $\pi_2|_{G_0}$ are equivalent. Does there exist an amalgam of these representations? Does this problem have a positive answer when the groups and spaces in question are finite, resp. finite-dimensional?*

In transcription, Question 4.5 asks: let G_1 and G_2 be two groups with a common subgroup G_0 . Let π_1 and π_2 be unitary representations of G_1 , respectively G_2 , on Hilbert spaces H_1 , respectively H_2 , with a common subspace H_0 , on which the restrictions $\pi_1|_{G_0}$ and $\pi_2|_{G_0}$ are equivalent. Does there exist an amalgam of these representations? Does this have a positive answer when the groups and spaces are finite, respectively finite-dimensional?

Definitions

We use the following standard interpretation of an amalgam of unitary representations. After replacing one copy of the common subrepresentation by an equivalent one, assume that there

is a Hilbert space H_0 , a unitary representation $\pi_0: G_0 \rightarrow U(H_0)$, and isometric G_0 -equivariant inclusions

$$u_i: H_0 \longrightarrow H_i, \quad u_i \pi_0(g) = \pi_i(g) u_i \quad (g \in G_0, i = 1, 2).$$

An amalgam consists of a group Γ containing G_1 and G_2 over their common G_0 , a Hilbert space K , a unitary representation $\rho: \Gamma \rightarrow U(K)$, and isometric maps $J_i: H_i \rightarrow K$ such that

$$J_1 u_1 = J_2 u_2, \quad \rho(g) J_i = J_i \pi_i(g) \quad (g \in G_i, i = 1, 2).$$

Idea of the proof

The source paper amalgamates metric actions by passing to a free product with amalgamation. For unitary representations, the parallel construction is not a metric graph but a tree of Hilbert spaces. The Bass-Serre tree of $G_1 *_{G_0} G_2$ has vertices of type G_1 and G_2 , and every edge remembers a coset of G_0 . Put a copy of H_i at each vertex of type G_i , put a copy of H_0 on each edge, and attach it to the two endpoint vertex spaces using the given G_0 -subrepresentation maps. Because the underlying graph is a tree, the Hilbert-space pushout is built by repeatedly adding only orthogonal complements; there is no cyclic compatibility condition. The left action of $G_1 *_{G_0} G_2$ on its Bass-Serre tree acts by unitaries on this direct-limit Hilbert space and gives the required amalgam.

For finite-dimensional input data, there is a second, more economical argument. Choose a very large multiple of the regular representation of the finite group G_0 . It is large enough to contain the common H_0 , the G_0 -orthogonal complement of H_0 in H_1 , and the corresponding complement in H_2 . The same G_0 -representation is also the restriction of large regular representations of G_1 and G_2 , and those regular representations can be chosen large enough to contain the original G_i -representations. Transporting these two G_i -actions back to the same G_0 -Hilbert space gives two finite-dimensional actions that agree on all of G_0 , so the free product with amalgamation acts on that finite space.

Lemma 1 (Hilbert direct limit over a tree). *Let T be a tree. For each vertex v , let E_v be a Hilbert space. For each edge $e = vw$, let E_e be a Hilbert space and let $\alpha_{e,v}: E_e \rightarrow E_v$ and $\alpha_{e,w}: E_e \rightarrow E_w$ be isometric linear embeddings. Then there are a Hilbert space K and isometric maps $J_v: E_v \rightarrow K$ such that*

$$J_v \alpha_{e,v} = J_w \alpha_{e,w} \quad (e = vw),$$

and K is generated by the subspaces $J_v(E_v)$.

Moreover, if a group G acts on this tree of Hilbert spaces by unitaries, then the action induces a unitary representation of G on K .

Proof. Root the tree at a vertex r . For each non-root vertex v , let $p(v)$ be its parent and let e_v be the edge joining v to $p(v)$. Let

$$L_v = E_v \ominus \alpha_{e_v,v}(E_{e_v}).$$

Set

$$K = E_r \oplus \bigoplus_{v \neq r} L_v.$$

Define J_r to be the inclusion of E_r . Suppose $J_{p(v)}$ has already been defined. Each $x \in E_v$ has a unique orthogonal decomposition

$$x = \alpha_{e_v,v}(a) + \ell, \quad a \in E_{e_v}, \ell \in L_v.$$

Set

$$J_v(x) = J_{p(v)}\alpha_{e_v, p(v)}(a) + \ell,$$

where ℓ is placed in the L_v -summand of K . This is an isometry, because the two displayed summands are orthogonal and $\alpha_{e_v, v}$, $\alpha_{e_v, p(v)}$ are isometries. Recursion along the unique paths from the root defines all J_v , and every edge is a parent-child edge for this rooting, so the required agreement holds.

For the equivariance statement, let G act on the data by tree automorphisms and fiber unitaries. On the algebraic span of the vertex images define

$$U_g J_v(x) = J_{gv}(gx).$$

The construction above gives the Hilbert direct limit of the tree diagram: inner products are forced by the edge identifications and by the orthogonal complement additions along unique paths. Since g preserves the tree, edge spaces, and all attaching maps by unitaries, it preserves those inner products. Therefore U_g is a well-defined isometry on a dense subspace of K , hence extends to a unitary operator. The formula immediately gives $U_g U_h = U_{gh}$. $\square$

Theorem 1. *Let G_1 and G_2 be groups with common subgroup G_0 . Let $\pi_i: G_i \rightarrow U(H_i)$, $i = 1, 2$, be unitary representations whose restrictions to G_0 have a common subrepresentation H_0 in the sense above. Then the two representations have an amalgam. In fact, they amalgamate inside a unitary representation of the free product with amalgamation*

$$\Gamma = G_1 *_{G_0} G_2.$$

Proof. Let T be the Bass-Serre tree of the amalgamated free product $\Gamma = G_1 *_{G_0} G_2$. Its vertices are

$$V(T) = \Gamma/G_1 \sqcup \Gamma/G_2,$$

and its edges are the cosets Γ/G_0 , with the edge γG_0 joining the vertices γG_1 and γG_2 .

We now define a Γ -equivariant tree of Hilbert spaces over T . For a vertex of type i , write

$$E_{\gamma G_i} = \Gamma \times_{G_i} H_i \quad \text{over the coset } \gamma G_i,$$

meaning that the fiber is represented by classes $[\gamma, \xi]_i$ with

$$[\gamma s, \xi]_i = [\gamma, \pi_i(s)\xi]_i \quad (s \in G_i).$$

For an edge γG_0 , set similarly

$$E_{\gamma G_0} = \Gamma \times_{G_0} H_0, \quad [\gamma s, h]_0 = [\gamma, \pi_0(s)h]_0 \quad (s \in G_0).$$

The edge maps into the two endpoint vertex fibers are

$$[\gamma, h]_0 \mapsto [\gamma, u_i h]_i, \quad i = 1, 2.$$

They are well-defined because the maps u_i intertwine the G_0 -actions: if $s \in G_0$, then

$$[\gamma s, u_i h]_i = [\gamma, \pi_i(s)u_i h]_i = [\gamma, u_i \pi_0(s)h]_i.$$

All edge maps are isometries.

Left multiplication by Γ on the first coordinate gives an equivariant action on the Bass-Serre tree and on this tree of Hilbert spaces. Applying Lemma 1, obtain a Hilbert space K , vertex embeddings J_v , and a unitary representation $\rho: \Gamma \rightarrow U(K)$.

Let $v_i = G_i$ be the base vertex of type i . Define

$$J_i: H_i \rightarrow K, \quad J_i(\xi) = J_{v_i}([1, \xi]_i).$$

The edge G_0 between G_1 and G_2 gives

$$J_1 u_1(h) = J_2 u_2(h) \quad (h \in H_0).$$

Finally, if $g \in G_i$, then g fixes the base vertex G_i , and our associated-bundle convention gives

$$\rho(g)J_i(\xi) = J_{v_i}([g, \xi]_i) = J_{v_i}([1, \pi_i(g)\xi]_i) = J_i\pi_i(g)\xi.$$

Thus $J_i(H_i)$ is G_i -invariant and carries exactly the original representation π_i . This is the desired amalgam. $\square$

Theorem 2 (finite-dimensional strengthening). *Assume, in addition to the hypotheses of Theorem 1, that G_0, G_1, G_2 are finite and that H_0, H_1, H_2 are finite-dimensional. Then the amalgam in Theorem 1 may be chosen on a finite-dimensional Hilbert space.*

Proof. Identify the two copies of H_0 , and write

$$H_1 = H_0 \oplus A, \quad H_2 = H_0 \oplus B$$

as orthogonal decompositions of G_0 -representations. The complements A and B are G_0 -invariant because the G_0 -actions are unitary. Let λ_0 denote the regular representation of G_0 . Choose an integer N so large that the G_0 -representation

$$M = N\lambda_0$$

contains $H_0 \oplus A \oplus B$ as a G_0 -subrepresentation. Increase N , if necessary, so that N is divisible by both indices $[G_1 : G_0]$ and $[G_2 : G_0]$, and so that

$$R_i = \frac{N}{[G_i : G_0]} \lambda_{G_i}$$

contains H_i as a G_i -subrepresentation for $i = 1, 2$. This is possible because every irreducible representation of a finite group occurs in its regular representation with positive multiplicity.

For each i , the restriction of R_i to G_0 is unitarily equivalent to M . Choose a G_0 -unitary isomorphism from the fixed space M onto $\text{Res}_{G_0}^{G_i} R_i$ which sends the fixed copy of H_0 to the distinguished H_0 inside the chosen H_i -copy, sends $H_0 \oplus A$ onto that chosen copy of H_1 when $i = 1$, and sends $H_0 \oplus B$ onto that chosen copy of H_2 when $i = 2$. Transport the G_i -action of R_i back to M , obtaining a representation $\rho_i: G_i \rightarrow U(M)$.

By construction, the two transported actions restrict to the same fixed G_0 -action on all of M . Also, the subspace $H_0 \oplus A \subseteq M$ is $\rho_1(G_1)$ -invariant and carries the original representation π_1 , while $H_0 \oplus B \subseteq M$ is $\rho_2(G_2)$ -invariant and carries π_2 . Since $\rho_1|_{G_0} = \rho_2|_{G_0}$, the universal property of $G_1 *_{G_0} G_2$ gives a unitary representation on the finite-dimensional Hilbert space M . The inclusions of H_1 and H_2 into M agree on H_0 , so this is the desired finite-dimensional amalgam. $\square$

Verification notes

The proof is purely structural and uses no finite search. The only standard group-theoretic input is the Bass-Serre tree of an amalgamated free product. The packet includes a short script, `code/verify_packet_assets.py`, which verifies that the source PDF contains the target question on page 21 and reproduces the screenshot crop.

The key point for review is Lemma 1: because the indexing graph is a tree, the Hilbert-space direct limit is obtained by successively adding orthogonal complements. If the indexing graph had cycles, additional compatibility conditions could appear; the Bass-Serre tree is precisely what removes that obstruction.

The key point for Theorem 2 is semisimplicity of finite group representations. A large regular representation of G_i contains any prescribed finite-dimensional G_i -representation, and its restriction to G_0 is a large multiple of the regular G_0 -representation. This lets the same fixed G_0 -Hilbert space support both transported finite-dimensional actions.

Limitations and novelty check

This is still marked as a partial packet relative to Doucha’s cluster of final open questions. It proves the unrestricted Hilbert-space amalgamation clause and also proves that finite-dimensional input data admit a finite-dimensional Hilbert-space amalgam. It does not prove that, for finite input groups, the image of $G_1 *_{G_0} G_2$ in the constructed finite-dimensional unitary group is finite. Thus if the final sentence of Question 4.5 is intended to ask for an amalgam inside the class of finite groups with finite-dimensional unitary representations, that stronger finite-image interpretation remains outside this packet.

On 2026-06-19, a bounded web search was run for the exact phrase “*Does there exist an amalgam of these representations*”, for “*Universal actions and representations of locally finite groups*” “*unitary representation*” amalgam, and for variants of *amalgamate unitary representations common subgroup*. No separate explicit answer was found in that search. This is evidence against an obvious duplicate, not a publication-level novelty guarantee.

Human review recommendation

Recommended verdict: likely valid strengthened partial result. A human reviewer should check the associated-bundle convention, the regular representation padding argument, and the intended meaning of the finite/fd sentence in Question 4.5. If finite image is not required, Theorem 2 appears to answer the finite-dimensional part. If finite image is required, a separate argument or counterexample is still needed.

References

- [1] M. Doucha, *Universal actions and representations of locally finite groups on metric spaces*, arXiv:1612.09448.
- [2] J.-P. Serre, *Trees*, Springer Monographs in Mathematics, Springer, 2003.